t3-w05 — Three dimensional shapes II — Three dimensional figures: properties of cylinders and spheres, and volume

Run `c0c3-deepseek-native` · conditions C0, C1, C2, C3 · the pack shown is draw 3 of each.

HER LESSON PLAN

Three dimensional shapes II — Three dimensional figures: properties of cylinders and spheres, and volume

Objectives

1. identify properties of cylinders and spheres
2. find the volume of a cube and a cuboid

Step 0 — intro: activating prior knowledge

Teacher recalls with the students what they have already been taught, so the work on cylinders, spheres and volume has something to stand on:

Step 1

Teacher displays shapes of cylinders and spheres while the students identify the properties. Teacher draws the shapes on the board and add to the properties already mentioned by the students if there is need.

- **teacher** displays shapes of cylinders and spheres
- **student** identify the properties
- **teacher** draws the shapes on the board and adds to the properties already mentioned by the students

1. 1. How many circular faces does a solid cylinder have? 2. A cylinder has how many faces.

Step II

Teacher explains how to derive the formula for volume of cube and cuboid while the students listen and ask questions where they do not understand. They copy worked examples into their notes, which also work the volume of a prism as area of base times height.

- **teacher** explains how to derive the formula for volume of cube and cuboid, and works the volume of a prism as area of base times height
- **student** listen and ask questions where they do not understand
- **student** copy worked examples into their notes

Evaluation

Teacher tests the students' understanding on the topic by giving them some questions to solve. Question: 1. Calculate the volume of a cuboid 8cm long, 4cm wide and 3cm high. 2. Calculate the volume of cuboid with dimensions 5m by 2m by 3m. 3. What is the volume of a cube of 4cm $\times$ 4cm $\times$ 4cm. 4. A rectangular tank hold 6m long by 2m wide holds 36m^{3} of water. How deep is the water in the tank.

1. Calculate the volume of a cuboid 8cm long, 4cm wide and 3cm high.
2. Calculate the volume of cuboid with dimensions 5m by 2m by 3m.
3. What is the volume of a cube of 4cm $\times$ 4cm $\times$ 4cm.
4. A rectangular tank hold 6m long by 2m wide holds 36m^{3} of water. How deep is the water in the tank.

HER LESSON NOTE

Verbatim, as transcribed. **C2 and C3 receive this; C0 and C1 never do** — which is the whole of what separates them.

Step 0 — intro: activating prior knowledge

```
1. properties of cubes and cuboid
2. properties of pyramids and cones
```

Step 1

```
#### Properties of a cylinder

**fig-01.** A cylinder drawn upright: an ellipse for the top face, two vertical sides
falling from its ends, and a curve closing the base.

\begin{center}
\begin{tikzpicture}[scale=0.55,line width=0.5pt]
  \draw (-1.2,2.6) arc (180:-180:1.2 and 0.35);
  \draw (-1.2,2.6) -- (-1.2,0);
  \draw (1.2,2.6) -- (1.2,0);
  \draw (-1.2,0) arc (180:360:1.2 and 0.35);
  \draw[dashed] (-1.2,0) arc (180:0:1.2 and 0.35);
\end{tikzpicture}
\end{center}

1. It has two circular faces and one curved face
2. It has no vertex
3. It has two curved edges
4. It has three faces altogether

#### Properties of sphere

**fig-02.** A sphere drawn as a circle with an ellipse across its middle to show the
curved surface.

\begin{center}
\begin{tikzpicture}[scale=0.55,line width=0.5pt]
  \draw (0,0) circle (1.3);
  \draw (-1.3,0) arc (180:360:1.3 and 0.42);
  \draw[dashed] (-1.3,0) arc (180:0:1.3 and 0.42);
\end{tikzpicture}
\end{center}

1. A sphere is symmetrical from all direction
2. It has no edge and no vertex
3. It has one curved surface

#### Student activity

1. How many circular faces does a solid cylinder have?
2. A cylinder has how many faces.
```

Step II

```
#### Volume of cube and cuboid

1. Volume of cuboid = Area of base $\times$ height

   $= \text{length} \times \text{breadth} \times \text{height}$

   $= lbh$

2. Volume of cube = Length $\times$ Length $\times$ Length

   $= l^{3}$

**Example 1.** Calculate the volume of a cuboid of length 3cm, breadth 2cm and height 8cm.

Solution:

Volume of cuboid $= L \times b \times h$

length $= 3cm$

breadth $= 2cm$

height $= 8cm$

Volume $= 3cm \times 2cm \times 8cm$

$= 48cm^{3}$

**Example 2.** The volume of a cuboid is 24cm$^{3}$. If the base is 6cm by 2cm. Find the
height of the cuboid.

Solution:

Volume of cuboid $= L \times b \times h$

length $= 6cm$

breadth $= 2cm$

height $= ?$

Volume $= 24cm^{3}$

$24cm^{3} = 6cm \times 2cm \times h$

$24cm^{3} = 12cm^{2}h$

Divide both side by 12cm$^{2}$

$\frac{24cm^{3}}{12cm^{2}} = \frac{12cm^{2}h}{12cm^{2}}$

$2cm = h$

$h = 2cm$

#### Volume of cube

**Example 3.** Calculate the volume of a cube of side 3cm.
```

```
Solution

Volume of cube $= l^{3}$

$V = (3cm)^{3}$

$V = 3cm \times 3cm \times 3cm$

$V = 27cm^{3}$

#### Volume of prism

Calculate the volume of the prism below.

**fig-03.** A triangular prism lying on its rectangular face. The triangular end has a base
of 5 m and a height of 3 m, and the prism runs back 10 m.

\begin{center}
\begin{tikzpicture}[scale=0.42,line width=0.5pt]
  \draw (0,0) -- (5,0) -- (2.5,3) -- cycle;
  \draw (5,0) -- (8.5,1.8);
  \draw (2.5,3) -- (6,4.8);
  \draw (8.5,1.8) -- (6,4.8);
  \draw[dashed] (0,0) -- (3.5,1.8);
  \draw[dashed] (3.5,1.8) -- (8.5,1.8);
  \draw[dashed] (3.5,1.8) -- (6,4.8);
  \draw[dashed] (2.5,0) -- (2.5,3);
  \node[below] at (2.5,-0.15) {$5\,\mathrm{m}$};
  \node[left] at (2.35,1.5) {$3\,\mathrm{m}$};
  \node[below right] at (6.9,0.8) {$10\,\mathrm{m}$};
\end{tikzpicture}
\end{center}

Solution:

Area of triangular face $= \frac{1}{2}bh$

$= \frac{1}{2} \times 5 \times 3$

$= 7.5m^{2}$

Volume of prism $=$ Area of base $\times$ height

$= 7.5m^{2} \times 10m$

$= 75m^{3}$
```

C0 `The curriculum objective alone. No lesson plan, no note.`

Input — what the model was sent

Contract `ComposeWriteObjectiveSection`, declaring `artefact_spec, lesson_context, unit_context`.

Lesson context — shared by every section

`subject`

Mathematics

`grade_level`

JSS1

`topic`

Three dimensional shapes II

`subtopic`

Three dimensional figures: properties of cylinders and spheres, and volume

`term`

Third Term

`week`

5

`lesson_objectives`

• identify properties of cylinders and spheres
• find the volume of a cube and a cuboid

Objective 1 — identify properties of cylinders and spheres

`objective_text`

identify properties of cylinders and spheres

Withheld from this payload: `step_title, step_content, instructional_type, estimated_minutes, current_step, instructional_events, student_tasks, activities, assessment_points, source_evidence, source_figures, objective_learner_state`.

Objective 2 — find the volume of a cube and a cuboid

`objective_text`

find the volume of a cube and a cuboid

Withheld from this payload: `step_title, step_content, instructional_type, estimated_minutes, current_step, instructional_events, student_tasks, activities, assessment_points, source_evidence, source_figures, objective_learner_state`.

Output — the pack it produced (draw 3)

Objective 1

Cylinders and Spheres: The Round Solids

> EXPLANATION
> Not every solid is built from flat faces. Two important solids that are round are the cylinder and the sphere. Being able to recognise them and say what they are made of helps you understand the world around you — from drinks tins to footballs.
>
> A cylinder has two flat circular faces (one on top and one underneath) and one curved surface that joins them. The two circular faces are the same size and lie exactly opposite each other. A tin of baked beans, a battery, or a drinking glass are everyday cylinders.
>
> A sphere is perfectly round, like a ball. It has no flat faces, no edges, and no corners. Every point on its surface is the same distance from its centre. A football, a marble, or an orange are spheres.
>
> The easiest way to tell a cylinder from a sphere is to count the flat faces and edges: a cylinder has two flat circular faces and a curved surface, while a sphere has no flat faces at all.

> WORKED_EXAMPLE
> Example 1: Is a soup tin a cylinder or a sphere?
>
> A soup tin has a flat circular top and a flat circular bottom, and a curved wall joining them. Because it has two flat circular faces and one curved surface, it is a cylinder.
>
> Example 2: How many flat faces does a sphere have?
>
> A sphere is perfectly round. It has no flat faces, no edges and no corners. So the answer is zero.
>
> Example 3: A drum has a circular top, a circular bottom, and a curved side. What solid is it, and how many curved surfaces does it have?
>
> It is a cylinder, because it has two circular faces and a curved side. It has exactly one curved surface.

1. State whether each of these is a cylinder or a sphere: (a) a football (b) a candle (c) an orange
answer: (a) sphere, (b) cylinder, (c) sphere

2. How many flat faces does a cylinder have, and what shape are they?
answer: A cylinder has 2 flat faces, and they are circles.

3. A solid has no flat faces, no edges and no corners, and every point on its surface is the same distance from its centre. What solid is it?
answer: It is a sphere.

Objective 2

Finding the Volume of Cubes and Cuboids

EXPLANATION

Volume is the amount of space a solid shape takes up. We measure volume in cubic units, such as cubic centimetres (cm^3) or cubic metres (m^3). A cube is a solid shape in which every edge is the same length, so it looks like a box with six equal square faces. A cuboid is a box too, but its length, width and height can all be different. To find how much space either one holds, we multiply three measurements together: length × width × height.

For a cuboid, call its three edges length, breadth (width) and height. Then:

Volume of a cuboid = length × breadth × height

For a cube, all three edges are equal to the same length. So instead of multiplying three different numbers, you multiply the one edge length by itself three times:

Volume of a cube = side × side × side, or $side^3$

Remember that the units of the answer are always cubic units, because you are multiplying three lengths together.

WORKED_EXAMPLE

Worked example 1 — volume of a cuboid

A cuboid measures 8 cm long, 5 cm broad and 3 cm high. Find its volume.

Use the rule: volume = length × breadth × height.

Volume = 8 cm × 5 cm × 3 cm
= 40 cm^2 × 3 cm
= 120 cm^3

So the volume of the cuboid is 120 cm^3.

WORKED_EXAMPLE

Worked example 2 — volume of a cube

A cube has an edge of length 4 cm. Find its volume.

Use the rule: volume = side × side × side.

Volume = 4 cm × 4 cm × 4 cm
= 16 cm^2 × 4 cm
= 64 cm^3

So the volume of the cube is 64 cm^3.

1. A cuboid has a length of 6 cm, a breadth of 4 cm and a height of 5 cm. What is its volume?
answer: 120 cm^3

2. A cube has an edge of length 5 m. What is its volume?
answer: 125 m^3

3. A box for storing books is a cuboid 30 cm long, 20 cm broad and 10 cm high. How much space does the box hold?
answer: 6000 cm^3

C1 `The objective plus the lesson plan's design: position, method labels, materials, entry behaviour and assessment.`

Input — what the model was sent

Contract `ComposeWriteObjectiveSection`, declaring `artefact_spec, lesson_context, unit_context`.

Lesson context — shared by every section

`subject`

Mathematics

`grade_level`

JSS1

`topic`

Three dimensional shapes II

`subtopic`

Three dimensional figures: properties of cylinders and spheres, and volume

`term`

Third Term

`week`

5

`lesson_objectives`

• identify properties of cylinders and spheres
• find the volume of a cube and a cuboid

`entry_behaviour`

• properties of cubes and cuboid
• properties of pyramids and cones

`instructional_materials`

• shapes of cylinders and spheres.

Objective 1 — identify properties of cylinders and spheres

`objective_text`

identify properties of cylinders and spheres

`current_step`

1

`instructional_events`

• present_content
• elicit_performance
• present_content

`student_tasks`

• 1. How many circular faces does a solid cylinder have? 2. A cylinder has how many faces.

Withheld from this payload: `step_title`, `step_content`, `instructional_type`, `estimated_minutes`, `activities`, `assessment_points`, `source_evidence`, `source_figures`, `objective_learner_state`.

Objective 2 — find the volume of a cube and a cuboid

`objective_text`

find the volume of a cube and a cuboid

`current_step`

2

`instructional_events`

• present_content
• provide_learning_guidance

`assessment_points`

• Calculate the volume of a cuboid 8cm long, 4cm wide and 3cm high.
• Calculate the volume of cuboid with dimensions 5m by 2m by 3m.
• What is the volume of a cube of 4cm $\times$ 4cm $\times$ 4cm.
• A rectangular tank hold 6m long by 2m wide holds 36m^{3} of water. How deep is the water in the tank.

Withheld from this payload: `step_title`, `step_content`, `instructional_type`, `estimated_minutes`, `student_tasks`, `activities`, `source_evidence`, `source_figures`, `objective_learner_state`.

Output — the pack it produced (draw 3)

Objective 1

Cylinders and Spheres: What to Look For

EXPLANATION

In this section you are going to look closely at two solid shapes you meet every day: a cylinder and a sphere. A solid shape is a three-dimensional figure, so it has length, width and height. Your job is to be able to describe what makes a cylinder a cylinder and a sphere a sphere — that is, to name their properties.

Start with a cylinder. Think of a tin of baked beans or a drinking cup that is straight up and down. A solid cylinder has two flat circular faces — one at the top and one at the bottom — and these two circles are exactly the same size. Between them is a curved surface that wraps around the sides, like the label on a tin. When you count the faces of a solid cylinder, you find the two flat circles plus the curved surface, so a solid cylinder has three faces in total. A cylinder also has two edges — the circular rims where the flat face meets the curved surface — and it has no vertices (no sharp corners), because the curved surface never comes to a point.

EXPLANATION

Now think of a sphere. A football, an orange or a ball bearing is a sphere. A sphere has just one face, and that face is completely curved — there is not a single flat part on it. Because its surface is smooth and curved all over, a sphere has no edges and no vertices at all. No matter which way you look at a sphere, or from which direction you measure it, it is the same shape all around — it is perfectly round.

To tell the two shapes apart, ask yourself about faces. A solid cylinder has two flat circular faces and a curved side, making three faces in total. A sphere has one curved face only. That is the key difference between them.

WORKED_EXAMPLE

Let's work through how to describe a solid cylinder, step by step, so you can see what to count.

Question: How many faces does a solid cylinder have?

Step 1 — Look for the flat faces. A solid cylinder has a flat circular face at the top and a flat circular face at the bottom. Those are two flat faces, and the two circles are the same size.

Step 2 — Look for any curved surface. The sides of the cylinder are one curved surface, like the wall of a tin. That curved surface counts as one more face.

Step 3 — Add them together. Two flat faces plus one curved face gives three faces in total.

So a solid cylinder has three faces: two circular faces and one curved face.

Now try the same counting for a sphere: there is no flat face anywhere, only one smooth curved surface. So a sphere has just one face.

1. 1. How many circular faces does a solid cylinder have?
answer: Two. A solid cylinder has a flat circular face at the top and a flat circular face at the bottom.

2. 2. How many faces does a solid cylinder have altogether?
answer: Three: two flat circular faces and one curved surface around the sides.

3. 3. How many flat faces does a sphere have?
answer: None. A sphere has one face only, and that face is completely curved — there is no flat surface anywhere on it.

Objective 2

Finding the Volume of a Cube and a Cuboid

EXPLANATION

Every three-dimensional shape takes up space, and the amount of space it takes up is called its volume. You already know the faces and edges of a cube and a cuboid; now you will find how much space each one holds.

Think of a cuboid as a box. A cube is a special cuboid whose length, width and height are all equal. To find the space a cuboid fills, picture it packed with small unit cubes (each one 1 cm by 1 cm by 1 cm, taking up 1 cubic centimetre, written 1 cm^3).

If a cuboid is length (l) long, width (w) wide and height (h) high, then its volume is the product of all three:

Volume of a cuboid = length × width × height = l × w × h

Why? Along the length you can fit l cubes in a row, along the width you can fit w of those rows in the base, and along the height you can stack h of those layers on top of each other. Multiplying them counts every cube in the shape.

Since a cube has all edges equal, we call each edge the side, s. Then length, width and height are all s, so:

Volume of a cube = s × s × s = s^3

WORKED_EXAMPLE

Worked example 1 – a cuboid
A cuboid is 8 cm long, 4 cm wide and 3 cm high. Find its volume.

Volume = length × width × height
= 8 cm × 4 cm × 3 cm
= 96 cm^3

The cuboid fills 96 cubic centimetres of space.

Worked example 2 – a cube
A cube has sides of 4 cm. Find its volume.

Volume = s × s × s
= 4 cm × 4 cm × 4 cm
= 64 cm^3

The cube fills 64 cubic centimetres of space.

Notice the units: when you multiply three lengths measured in centimetres, the volume is in cubic centimetres (cm^3). If the lengths are in metres, the volume is in cubic metres (m^3).

WORKED_EXAMPLE

Worked example 3 – finding a missing side from the volume
A rectangular tank is 6 m long and 2 m wide. It holds 36 m^3 of water. How deep is the water?

The tank is a cuboid, so volume = length × width × depth. We know the volume, the length and the width, and we want the depth.

Volume = length × width × depth
36 = 6 × 2 × depth
36 = 12 × depth

To undo the multiplying by 12, divide both sides by 12:
depth = 36 ÷ 12 = 3 m

The water is 3 m deep.

1. Calculate the volume of a cuboid that is 8 cm long, 4 cm wide and 3 cm high.

answer: 96 cm^3

2. Calculate the volume of a cuboid with dimensions 5 m by 2 m by 3 m.

answer: 30 m^3

3. What is the volume of a cube measuring 4 cm × 4 cm × 4 cm?

answer: 64 cm^3

4. A rectangular tank that is 6 m long and 2 m wide holds 36 m^3 of water. How deep is the water in the tank?

answer: 3 m deep

C2 `C1 plus the lesson note as source evidence.`

Input — what the model was sent

Contract `ComposeWriteGroundedSection`, declaring `artefact_spec, lesson_context, unit_context, lesson_content`.

Lesson context — shared by every section

`subject`

Mathematics

`grade_level`

JSS1

`topic`

Three dimensional shapes II

`subtopic`

Three dimensional figures: properties of cylinders and spheres, and volume

`term`

Third Term

`week`

5

`lesson_objectives`

• identify properties of cylinders and spheres
• find the volume of a cube and a cuboid

`entry_behaviour`

• properties of cubes and cuboid
• properties of pyramids and cones

`instructional_materials`

• shapes of cylinders and spheres.

Objective 1 — identify properties of cylinders and spheres

objective_text

identify properties of cylinders and spheres

current_step

1

instructional_events

• present_content
• elicit_performance
• present_content

student_tasks

• 1. How many circular faces does a solid cylinder have? 2. A cylinder has how many faces.

source_figures


```
[
 {
  "caption": "A cylinder drawn upright: an ellipse for the top face, two vertical sides falling
from its ends, and a curve closing the base.",
  "figure_ref": "fig-01"
 },
 {
  "caption": "A sphere drawn as a circle with an ellipse across its middle to show the curved
surface.",
  "figure_ref": "fig-02"
 }
]
```


lesson_content


```
[
 {
  "source_ref": "the lesson note",
  "excerpt": "#### Properties of a cylinder\n\n**fig-01.** A cylinder drawn upright: an ellipse
for the top face, two vertical sides\nfalling from its ends, and a curve closing the base.
\n\n[drawing attached to this section as fig-01]\n\n1. It has two circular faces and one curved
face\n2. It has no vertex\n3. It has two curved edges\n4. It has three faces altogether\n\n####
Properties of sphere\n\n**fig-02.** A sphere drawn as a circle with an ellipse across its middle
to show the\ncurved surface.\n\n[drawing attached to this section as fig-02]\n\n1. A sphere is
symmetrical from all direction\n2. It has no edge and no vertex\n3. It has one curved
surface\n\n#### Student activity\n\n1. How many circular faces does a solid cylinder have?\n2. A
cylinder has how many faces."
 }
]
```


Withheld from this payload: `step_title`, `step_content`, `instructional_type`, `estimated_minutes`, `activities`, `assessment_points`, `source_evidence`, `objective_learner_state`.

Objective 2 — find the volume of a cube and a cuboid

objective_text

find the volume of a cube and a cuboid

current_step

2

instructional_events

• present_content
• provide_learning_guidance

assessment_points

• Calculate the volume of a cuboid 8cm long, 4cm wide and 3cm high.
• Calculate the volume of cuboid with dimensions 5m by 2m by 3m.
• What is the volume of a cube of 4cm $\times$ 4cm $\times$ 4cm.
• A rectangular tank hold 6m long by 2m wide holds 36m^{3} of water. How deep is the water in the tank.

source_figures


```
[
 {
  "caption": "A triangular prism lying on its rectangular face. The triangular end has a base of
5 m and a height of 3 m, and the prism runs back 10 m.",
  "figure_ref": "fig-03"
 }
]
```


lesson_content


```
[
 {
  "source_ref": "the lesson note",
  "excerpt": "#### Volume of cube and cuboid\n\n1. Volume of cuboid = Area of base $\\times$
height\n\n   $= \\text{length} \\times \\text{breadth} \\times \\text{height}$\n\n   $= lbh$
\n\n2. Volume of cube = Length $\\times$ Length $\\times$ Length\n\n   $= l^{3}$\n\n**Example
1.** Calculate the volume of a cuboid of length 3cm, breadth 2cm and height 8cm.\n\nSolution:
\n\nVolume of cuboid $= L \\times b \\times h$\n\nlength $= 3cm$\n\nbreadth $= 2cm$\n\nheight $=
8cm$\n\nVolume $= 3cm \\times 2cm \\times 8cm$\n\n$= 48cm^{3}$\n\n**Example 2.** The volume of a
cuboid is 24cm$^{3}$. If the base is 6cm by 2cm. Find the\nheight of the cuboid.\n\nSolution:
\n\nVolume of cuboid $= L \\times b \\times h$\n\nlength $= 6cm$\n\nbreadth $= 2cm$\n\nheight
$= ?$\n\nVolume $= 24cm^{3}$\n\n$24cm^{3} = 6cm \\times 2cm \\times h$\n\n$24cm^{3} = 12cm^{2}h$
\n\nDivide both side by 12cm$^{2}$\n\n$\\frac{24cm^{3}}{12cm^{2}} = \\frac{12cm^{2}h}{12cm^{2}}$
\n\n$2cm = h$\n\n$h = 2cm$\n\n#### Volume of cube\n\n**Example 3.** Calculate the volume of a
cube of side 3cm.\n\nSolution\n\nVolume of cube $= l^{3}$\n\n$V = (3cm)^{3}$\n\n$V = 3cm \\times
3cm \\times 3cm$\n\n$V = 27cm^{3}$\n\n#### Volume of prism\n\nCalculate the volume of the prism
below.\n\n**fig-03.** A triangular prism lying on its rectangular face. The triangular end has a
base\nof 5 m and a height of 3 m, and the prism runs back 10 m.\n\n[drawing attached to this
section as fig-03, marked 5 m, 3 m, 10 m]\n\nSolution:\n\nArea of triangular face $= \\frac{1}{2}
bh$\n\n$= \\frac{1}{2} \\times 5 \\times 3$\n\n$= 7.5m^{2}$\n\nVolume of prism $=$ Area of base $
\\times$ height\n\n$= 7.5m^{2} \\times 10m$\n\n$= 75m^{3}$"
 }
]
```

Withheld from this payload: `step_title, step_content, instructional_type, estimated_minutes, student_tasks, activities, source_evidence, objective_learner_state`.

Output — the pack it produced (draw 3)

Objective 1

Properties of Cylinders and Spheres

> EXPLANATION
> You already know your way around a cube, a cuboid, and shapes like pyramids and cones. Now let's meet two more solid shapes and learn their properties: the cylinder and the sphere. These are three-dimensional figures, so when we describe them we look at their faces, their edges, and their vertices (corners).

> EXPLANATION
> Look at fig-01, a solid cylinder drawn upright. The top of the cylinder is a circle, and the bottom is a circle too — those are its two circular faces. Between them runs a smooth curved face that wraps all the way around, so the cylinder has one curved face. Put that together and the cylinder has three faces altogether: two flat circular faces and one curved face.
>
> Where does that curved face meet each circular face? Along a circle — and since it meets both the top and the bottom, the cylinder has two curved edges. And because a cylinder is round with no pointy corners, it has no vertex at all.

> VISUAL
> ```
> \begin{tikzpicture}[scale=0.55,line width=0.5pt]
>  \draw (-1.2,2.6) arc (180:-180:1.2 and 0.35);
>  \draw (-1.2,2.6) -- (-1.2,0);
>  \draw (1.2,2.6) -- (1.2,0);
>  \draw (-1.2,0) arc (180:360:1.2 and 0.35);
>  \draw[dashed] (-1.2,0) arc (180:0:1.2 and 0.35);
> \end{tikzpicture}
> ```

> EXPLANATION
> Now look at fig-02, a sphere. A sphere is the shape of a ball — every point on its surface is the same distance from its centre. Because of that, a sphere is symmetrical from every direction: turn it any way you like and it looks the same.
>
> A sphere has exactly one curved surface. It has no flat faces, no edges, and no vertices — roll it around and there is nothing sharp or flat about it at all.

> VISUAL
> ```
> \begin{tikzpicture}[scale=0.55,line width=0.5pt]
>  \draw (0,0) circle (1.3);
>  \draw (-1.3,0) arc (180:360:1.3 and 0.42);
>  \draw[dashed] (-1.3,0) arc (180:0:1.3 and 0.42);
> \end{tikzpicture}
> ```

> WORKED_EXAMPLE
> Let's collect what we have found.
>
> A cylinder has:
> - two circular faces and one curved face (three faces altogether),
> - no vertex,
> - two curved edges.
>
> A sphere has:
> - one curved surface,
> - no edge and no vertex,
> - and it is symmetrical from every direction.

1. 1. How many circular faces does a solid cylinder have?
   answer: Two — the top circle and the bottom circle.
2. 2. A solid cylinder has how many faces altogether?
   answer: Three — two circular faces and one curved face.

Objective 2

Finding the Volume of a Cube and a Cuboid

EXPLANATION

Volume is the amount of space a three-dimensional shape takes up, and it is measured in cubic units such as cubic centimetres (cm^3) or cubic metres (m^3).

A cuboid is a box whose faces are rectangles. Its volume is found by multiplying its length, breadth and height together:

Volume of cuboid = length × breadth × height = lbh

You can also think of it as the area of the base times the height:

Volume of cuboid = Area of base × height

A cube is a special cuboid in which length, breadth and height are all equal. If the side of the cube is l, then:

Volume of cube = length × length × length = l × l × l = l^3

So once you know one side of a cube, cubing it gives the volume.

WORKED_EXAMPLE

Example 1. Calculate the volume of a cuboid of length 3 cm, breadth 2 cm and height 8 cm.

Volume of cuboid = L × b × h

length = 3 cm
breadth = 2 cm
height = 8 cm

Volume = 3 cm × 2 cm × 8 cm
= 48 cm^3

WORKED_EXAMPLE

Example 2. The volume of a cuboid is 24 cm^3. Its base is 6 cm by 2 cm. Find the height of the cuboid.

Volume of cuboid = L × b × h

length = 6 cm
breadth = 2 cm
height = ?

Volume = 24 cm^3

24 cm^3 = 6 cm × 2 cm × h
24 cm^3 = 12 cm^2 × h

Divide both sides by 12 cm^2:

24 cm^3 ÷ 12 cm^2 = h
2 cm = h

So the height of the cuboid is 2 cm.

WORKED_EXAMPLE

Example 3. Calculate the volume of a cube of side 3 cm.

Volume of cube = l^3

V = $(3\ cm)^3$

$V = 3\ \text{cm} \times 3\ \text{cm} \times 3\ \text{cm}$

$V = 27\ \text{cm}^3$

VISUAL

```
\begin{tikzpicture}[scale=0.42,line width=0.5pt]
  \draw (0,0) -- (5,0) -- (2.5,3) -- cycle;
  \draw (5,0) -- (8.5,1.8);
  \draw (2.5,3) -- (6,4.8);
  \draw (8.5,1.8) -- (6,4.8);
  \draw[dashed] (0,0) -- (3.5,1.8);
  \draw[dashed] (3.5,1.8) -- (8.5,1.8);
  \draw[dashed] (3.5,1.8) -- (6,4.8);
  \draw[dashed] (2.5,0) -- (2.5,3);
  \node[below] at (2.5,-0.15) {$5\,\mathrm{m}$};
  \node[left] at (2.35,1.5) {$3\,\mathrm{m}$};
  \node[below right] at (6.9,0.8) {$10\,\mathrm{m}$};
\end{tikzpicture}
```

1. Calculate the volume of a cuboid that is 8 cm long, 4 cm wide and 3 cm high.

answer: 96 cm^3

2. Calculate the volume of a cuboid with dimensions 5 m by 2 m by 3 m.

answer: 30 m^3

3. What is the volume of a cube of side 4 cm?

answer: 64 cm^3

4. A rectangular tank that is 6 m long and 2 m wide holds 36 m^3 of water. How deep is the water in the tank?

answer: 3 m

C3 `C2 plus a validation and repair pass over the draft.`

Input — what the model was sent

Contract `ComposeWriteGroundedSection`, declaring `artefact_spec, lesson_context, unit_context, lesson_content`.

Lesson context — shared by every section

`subject`

Mathematics

`grade_level`

JSS1

`topic`

Three dimensional shapes II

`subtopic`

Three dimensional figures: properties of cylinders and spheres, and volume

`term`

Third Term

`week`

5

`lesson_objectives`

• identify properties of cylinders and spheres
• find the volume of a cube and a cuboid

`entry_behaviour`

• properties of cubes and cuboid
• properties of pyramids and cones

`instructional_materials`

• shapes of cylinders and spheres.

Objective 1 — identify properties of cylinders and spheres

objective_text

identify properties of cylinders and spheres

current_step

1

instructional_events

• present_content
• elicit_performance
• present_content

student_tasks

• 1. How many circular faces does a solid cylinder have? 2. A cylinder has how many faces.

source_figures


```
[
 {
  "caption": "A cylinder drawn upright: an ellipse for the top face, two vertical sides falling
from its ends, and a curve closing the base.",
  "figure_ref": "fig-01"
 },
 {
  "caption": "A sphere drawn as a circle with an ellipse across its middle to show the curved
surface.",
  "figure_ref": "fig-02"
 }
]
```


lesson_content


```
[
 {
  "source_ref": "the lesson note",
  "excerpt": "#### Properties of a cylinder\n\n**fig-01.** A cylinder drawn upright: an ellipse
for the top face, two vertical sides\nfalling from its ends, and a curve closing the base.
\n\n[drawing attached to this section as fig-01]\n\n1. It has two circular faces and one curved
face\n2. It has no vertex\n3. It has two curved edges\n4. It has three faces altogether\n\n####
Properties of sphere\n\n**fig-02.** A sphere drawn as a circle with an ellipse across its middle
to show the\ncurved surface.\n\n[drawing attached to this section as fig-02]\n\n1. A sphere is
symmetrical from all direction\n2. It has no edge and no vertex\n3. It has one curved
surface\n\n#### Student activity\n\n1. How many circular faces does a solid cylinder have?\n2. A
cylinder has how many faces."
 }
]
```


Withheld from this payload: `step_title`, `step_content`, `instructional_type`, `estimated_minutes`, `activities`, `assessment_points`, `source_evidence`, `objective_learner_state`.

Objective 2 — find the volume of a cube and a cuboid

objective_text

find the volume of a cube and a cuboid

current_step

2

instructional_events

• present_content
• provide_learning_guidance

assessment_points

• Calculate the volume of a cuboid 8cm long, 4cm wide and 3cm high.
• Calculate the volume of cuboid with dimensions 5m by 2m by 3m.
• What is the volume of a cube of 4cm $\times$ 4cm $\times$ 4cm.
• A rectangular tank hold 6m long by 2m wide holds 36m^{3} of water. How deep is the water in the tank.

source_figures


```
[
 {
  "caption": "A triangular prism lying on its rectangular face. The triangular end has a base of
5 m and a height of 3 m, and the prism runs back 10 m.",
  "figure_ref": "fig-03"
 }
]
```


lesson_content


```
[
 {
  "source_ref": "the lesson note",
  "excerpt": "#### Volume of cube and cuboid\n\n1. Volume of cuboid = Area of base $\\times$
height\n\n   $= \\text{length} \\times \\text{breadth} \\times \\text{height}$\n\n   $= lbh$
\n\n2. Volume of cube = Length $\\times$ Length $\\times$ Length\n\n   $= l^{3}$\n\n**Example
1.** Calculate the volume of a cuboid of length 3cm, breadth 2cm and height 8cm.\n\nSolution:
\n\nVolume of cuboid $= L \\times b \\times h$\n\nlength $= 3cm$\n\nbreadth $= 2cm$\n\nheight $=
8cm$\n\nVolume $= 3cm \\times 2cm \\times 8cm$\n\n$= 48cm^{3}$\n\n**Example 2.** The volume of a
cuboid is 24cm$^{3}$. If the base is 6cm by 2cm. Find the\nheight of the cuboid.\n\nSolution:
\n\nVolume of cuboid $= L \\times b \\times h$\n\nlength $= 6cm$\n\nbreadth $= 2cm$\n\nheight
$= ?$\n\nVolume $= 24cm^{3}$\n\n$24cm^{3} = 6cm \\times 2cm \\times h$\n\n$24cm^{3} = 12cm^{2}h$
\n\nDivide both side by 12cm$^{2}$\n\n$\\frac{24cm^{3}}{12cm^{2}} = \\frac{12cm^{2}h}{12cm^{2}}$
\n\n$2cm = h$\n\n$h = 2cm$\n\n#### Volume of cube\n\n**Example 3.** Calculate the volume of a
cube of side 3cm.\n\nSolution\n\nVolume of cube $= l^{3}$\n\n$V = (3cm)^{3}$\n\n$V = 3cm \\times
3cm \\times 3cm$\n\n$V = 27cm^{3}$\n\n#### Volume of prism\n\nCalculate the volume of the prism
below.\n\n**fig-03.** A triangular prism lying on its rectangular face. The triangular end has a
base\nof 5 m and a height of 3 m, and the prism runs back 10 m.\n\n[drawing attached to this
section as fig-03, marked 5 m, 3 m, 10 m]\n\nSolution:\n\nArea of triangular face $= \\frac{1}{2}
bh$\n\n$= \\frac{1}{2} \\times 5 \\times 3$\n\n$= 7.5m^{2}$\n\nVolume of prism $=$ Area of base $
\\times$ height\n\n$= 7.5m^{2} \\times 10m$\n\n$= 75m^{3}$"
 }
]
```

Withheld from this payload: `step_title, step_content, instructional_type, estimated_minutes, student_tasks, activities, source_evidence, objective_learner_state`.

Output — the pack it produced (draw 3)

Objective 1

Cylinders and Spheres: Identifying Their Parts

> EXPLANATION
> Two common three-dimensional figures are the cylinder and the sphere. You meet them every day: a tin of milk is a cylinder, and a ball is a sphere. Let's look closely at the parts each one has.

> EXPLANATION
> Look at fig-01, the cylinder drawn upright. A cylinder has two circular faces — one at the top and one at the bottom — joined by a curved surface that runs around the sides. Because its faces are flat circles, a cylinder has no pointed corner where faces meet, so it has no vertex. The edge where each circular face meets the curved surface is curved, and a cylinder has two such curved edges.
>
> So a solid cylinder has:
> 1. Two circular faces and one curved face — three faces altogether.
> 2. No vertex.
> 3. Two curved edges.

> VISUAL
> ```
> \begin{tikzpicture}[scale=0.55,line width=0.5pt]
>  \draw (-1.2,2.6) arc (180:-180:1.2 and 0.35);
>  \draw (-1.2,2.6) -- (-1.2,0);
>  \draw (1.2,2.6) -- (1.2,0);
>  \draw (-1.2,0) arc (180:360:1.2 and 0.35);
>  \draw[dashed] (-1.2,0) arc (180:0:1.2 and 0.35);
> \end{tikzpicture}
> ```

> EXPLANATION
> Now look at fig-02, the sphere. A sphere is a perfectly round ball. Whichever way you turn it, it looks the same from every direction, so a sphere is symmetrical from all directions. Its whole surface is one curved surface, and it has no edge and no vertex.
>
> So a sphere has:
> 1. One curved surface.
> 2. No edge and no vertex.
> 3. It is symmetrical from all directions.

> VISUAL
> ```
> \begin{tikzpicture}[scale=0.55,line width=0.5pt]
>  \draw (0,0) circle (1.3);
>  \draw (-1.3,0) arc (180:360:1.3 and 0.42);
>  \draw[dashed] (-1.3,0) arc (180:0:1.3 and 0.42);
> \end{tikzpicture}
> ```

> WORKED_EXAMPLE
> Let's name the parts of a solid cylinder together. Imagine holding a tin of milk upright.
>
> How many circular faces does it have? There is a circular face at the top and a circular face at the bottom — that is two circular faces. The sides between them are one curved face, so the cylinder has three faces altogether.
>
> Does it have a vertex? The circular faces meet the curved surface in smooth curved edges, not in pointed corners, so a cylinder has no vertex.

1. How many circular faces does a solid cylinder have?
answer: A solid cylinder has two circular faces (one at the top and one at the bottom).

2. A cylinder has how many faces altogether?
answer: A cylinder has three faces altogether: two circular faces and one curved face.

3. Does a sphere have any edges or vertices? What surface does it have?
answer: A sphere has no edge and no vertex, and it has one curved surface.

Finding the Volume of Cubes and Cuboids

EXPLANATION

A cuboid is a box with six rectangular faces, like a brick or a shoebox. Its volume is the amount of space it fills, measured in cubic units such as cm^3 or m^3.

You can find a cuboid's volume by multiplying the area of its base by its height. The base is a rectangle with length L and breadth b, so its area is L × b. Multiply that by the height h:

Volume of cuboid = area of base × height
= length × breadth × height
= L × b × h

A cube is a special cuboid in which the length, breadth and height are all equal. If each side is l, then:

Volume of cube = l × l × l
= l^3

WORKED_EXAMPLE

Example 1. Calculate the volume of a cuboid of length 3 cm, breadth 2 cm and height 8 cm.

Solution.
Volume of cuboid = L × b × h

length = 3 cm, breadth = 2 cm, height = 8 cm

Volume = 3 cm × 2 cm × 8 cm
= 48 cm^3

So the cuboid holds 48 cubic centimetres.

WORKED_EXAMPLE

Example 2. The volume of a cuboid is 24 cm^3. Its base is 6 cm by 2 cm. Find its height.

Solution.
Volume of cuboid = L × b × h

length = 6 cm, breadth = 2 cm, height = ?

We know Volume = 24 cm^3, so:

24 cm^3 = 6 cm × 2 cm × h
24 cm^3 = 12 cm^2 × h

Divide both sides by 12 cm^2:

24 cm^3 ÷ 12 cm^2 = h
2 cm = h

So the height of the cuboid is 2 cm.

WORKED_EXAMPLE

Example 3. Calculate the volume of a cube of side 3 cm.

Solution.
Volume of cube = l^3

V = $(3\ cm)^3$
= 3 cm × 3 cm × 3 cm
= 27 cm^3

EXPLANATION

The same idea works for any prism: a shape with a constant cross-section running back a certain distance. A prism's volume is the area of its base (the end face) multiplied by its length.

Look at the triangular prism. Its triangular end has a base of 5 m and a height of 3 m, and the prism runs back 10 m.

Area of the triangular face = ½ × base × height
= ½ × 5 × 3
= 7.5 m^2

Volume of prism = area of base × length
= 7.5 m^2 × 10 m
= 75 m^3

VISUAL

```
\begin{tikzpicture}[scale=0.42,line width=0.5pt]
  \draw (0,0) -- (5,0) -- (2.5,3) -- cycle;
  \draw (5,0) -- (8.5,1.8);
  \draw (2.5,3) -- (6,4.8);
  \draw (8.5,1.8) -- (6,4.8);
  \draw[dashed] (0,0) -- (3.5,1.8);
  \draw[dashed] (3.5,1.8) -- (8.5,1.8);
  \draw[dashed] (3.5,1.8) -- (6,4.8);
  \draw[dashed] (2.5,0) -- (2.5,3);
  \node[below] at (2.5,-0.15) {$5\,\mathrm{m}$};
  \node[left] at (2.35,1.5) {$3\,\mathrm{m}$};
  \node[below right] at (6.9,0.8) {$10\,\mathrm{m}$};
\end{tikzpicture}
```

1. Calculate the volume of a cuboid that is 8 cm long, 4 cm wide and 3 cm high.
answer: 96 cm^3

2. Calculate the volume of a cuboid with dimensions 5 m by 2 m by 3 m.
answer: 30 m^3

3. What is the volume of a cube with side 4 cm?
answer: 64 cm^3

4. A rectangular tank is 6 m long and 2 m wide, and it holds 36 m^3 of water. How deep is the water in the tank?
answer: 3 m

Result — per rubric dimension

Scores are 1–5. Every figure is the mean of 3 draws rated by 2 judge pass(es) — 6 judgments per cell.

Rubric	C0	C1	C2	C3	C1 – C0	C2 – C1	C3 – C2
Objective and curriculum alignment	4.67	5.00	4.17	4.67	+0.33	-0.83	+0.50
Taught-lesson alignment	3.67	4.00	3.83	4.00	+0.33	-0.17	+0.17
Scope and conceptual correctness	4.33	4.83	3.00	3.50	+0.50	-1.83	+0.50
Student-facing clarity and usefulness	5.00	5.00	4.50	4.83	+0.00	-0.50	+0.33

Why the judge scored it that way

Every rationale the judge wrote for this lesson, deduplicated across draws and passes. These are its words, not a summary.

Objective and curriculum alignment — C0 4.67, C1 5.00, C2 4.17, C3 4.67

- `C0` The pack addresses both stated lesson objectives clearly and directly. It covers cylinder and sphere properties and cube/cuboid volume, matching what the teacher set out to teach and assess. A point is withheld because the lesson subtopic mentions 'volume' for cylinders and spheres as well, yet the pack (like the teacher's own evaluation) restricts volume work to cubes and cuboids — this is correct to the evaluation but leaves a minor ambiguity in full curriculum coverage.
- `C0` The revision pack addresses both lesson objectives fully and in the correct JSS1 Third Term context. Every topic the teacher set out to teach is represented clearly.
- `C0` The revision pack maps directly onto both stated lesson objectives with appropriate depth for JSS1. No objective is omitted or distorted.
- `C0` Both stated objectives are covered and the pack does not venture into unteaught objectives. The only minor gap is that the pack does not include any evaluation-style questions that mirror the teacher's own depth (e.g., the reverse-calculation tank problem), so alignment is strong but not perfect.
- `C0` The revision pack maps exactly onto both stated lesson objectives: identifying properties of cylinders and spheres, and finding the volume of a cube and cuboid. All key content expected by the JSS1 Third Term curriculum for this topic is present.
- `C0` Both lesson objectives are fully addressed in the revision pack. The pack covers the curriculum target for JSS1 Term 3 Week 5 with appropriate depth and content.
- `C1` The pack directly and completely covers both stated lesson objectives — properties of cylinders and spheres, and volume of cubes and cuboids — with no gaps and no significant drift outside the curriculum target for JSS1 Term 3 week 5.
- `C1` The revision pack maps directly onto both stated lesson objectives with no gaps. Every curriculum target from the lesson plan is represented in appropriate depth for JSS1.
- `C1` Both stated lesson objectives are fully addressed in the revision pack, and the content maps directly onto the JSS1 Third Term curriculum target for three-dimensional shapes. Nothing important is missing.
- `C1` The revision pack addresses every stated objective directly and completely, with no significant omissions or additions that fall outside the curriculum target for JSS1 Term 3 Week 5. Alignment is essentially perfect.
- `C1` The revision pack covers both stated lesson objectives completely and in the correct sequence. Nothing important from the curriculum target is missing, and no out-of-scope material is introduced.
- `C1` The pack covers both stated lesson objectives completely and in the correct JSS1 Third Term scope. Nothing essential from the objectives is missing.
- `C2` The pack aligns well with both stated objectives and the JSS1 curriculum target. It loses one point because the objectives list cube/cuboid volume but the pack also introduces triangular prism volume, which goes slightly beyond the stated objectives.
- `C2` The revision pack covers both stated objectives clearly and completely. The minor gap is that the subtopic hints at cylinder/sphere volume but the objectives don't require it, so this is not a real misalignment against the objectives as written.
- `C2` Both stated objectives are fully addressed in correct sequence and at the right curriculum level for JSS1. The pack does not stray outside the lesson scope.

- `C2` Both stated lesson objectives are addressed directly, and the practice questions match the teacher's own evaluation tasks. However, the subtopic heading promises volume of cylinders and spheres, which the pack never delivers, leaving a small but real gap in curriculum alignment.
- `C2` Both stated objectives are addressed clearly and in order. The pack matches the curriculum target for JSS1 Term 3 on 3D shapes. A minor gap is that the lesson objective says 'volume of a cube and a cuboid' but the pack adds a triangular prism example that goes slightly beyond the stated objectives, preventing a full 5.
- `C2` Both stated objectives are covered clearly and the curriculum content is appropriate for JSS1 Third Term. The pack loses a point because it never explicitly states or derives the volume of a cube as a special case of the cuboid in a way that ties back to the cube's equal sides (though it does present the formula), and there is a stray triangular prism figure that does not belong to the lesson objectives at all.
- `C3` Both lesson objectives are fully addressed in the correct order. The content matches the JSS1 Third Term curriculum target for 3D shapes, and nothing outside the stated objectives is examined in the practice questions.
- `C3` Both stated lesson objectives are addressed completely and in the correct order. The pack matches the JSS1 Third Term curriculum target for this topic without omission.
- `C3` Both lesson objectives are addressed clearly and in sequence. The only minor gap is that the lesson topic heading mentions 'volume' of cylinders/spheres but the objectives specify cubes/cuboids; the pack sensibly follows the objectives rather than the topic heading, which is appropriate but means the topic title is slightly mismatched.
- `C3` The revision pack covers both stated lesson objectives completely and accurately, aligning well with the JSS1 curriculum target for 3D shapes and volume.
- `C3` Both stated lesson objectives are fully addressed in dedicated sections with appropriate depth. The pack maps one-to-one onto the curriculum targets for this JSS1 Third Term week-5 lesson.
- `C3` Both stated objectives are addressed directly and in reasonable depth. The pack loses one point because the lesson's second objective says 'volume of a cube and a cuboid' only, yet the pack extends into the volume of a triangular prism, which is beyond the stated objectives for this lesson.

Taught-lesson alignment — C0 3.67, C1 4.00, C2 3.83, C3 4.00

- `C0` The pack follows the same topic sequence as the lesson and covers the right content, but it does not reflect the taught lesson's activity-based, student-participatory character, and it omits the reverse/application question type (finding depth) that featured in the teacher's evaluation, reducing alignment with the actual depth and method of the lesson.
- `C0` The pack faithfully reflects the lesson sequence, emphasis, and expected depth. The slight gap is that the lesson's interactive identification activity (students pointing out properties from displayed shapes) is naturally lost in a text-based pack, but the content coverage compensates well.
- `C0` The pack faithfully follows the lesson's sequence and depth, and its practice questions align well with the teacher's evaluation questions. Minor divergence is that the tactile/display activity from the presentation is naturally absent, which is acceptable for home revision.
- `C0` The pack follows the lesson sequence and covers the same two topics, but it does not fully reflect the depth the teacher demonstrated in the evaluation (especially the reverse-calculation question) and the interactive identification activity is replaced by expository text, so the emphasis and expected depth are only partially matched.
- `C0` The pack mirrors the lesson's order and main activities well. The only gap is that none of the teacher's own evaluation questions (including the reverse/depth problem) appear, so the specific assessment emphasis of the taught lesson is partially missed, but overall alignment is strong.
- `C0` The pack closely follows the lesson sequence and method, with worked examples clearly aligned to the teacher's approach. The omission of the inverse volume problem (finding depth) is a minor gap, but the overall alignment is strong.
- `C1` The pack faithfully reflects the lesson's sequence and depth, reproducing the teacher's own set questions and worked examples; the slight departure is that the tactile identification activity becomes descriptive text, which is appropriate for a take-home format but means the activity emphasis is not perfectly mirrored.
- `C1` The pack faithfully reflects the lesson's sequence and method, including the formula-derivation approach and the teacher's own evaluation questions. The only notable gap is that the lesson's explicit link to prior knowledge (cubes, cuboids, pyramids, cones) is not echoed in the revision material.
- `C1` The pack faithfully reproduces the lesson's sequence and emphasis; the only minor gap is that it does not replicate the hands-on display/identification activity with physical shapes, but that is appropriate for a take-home text format.
- `C1` The pack closely mirrors the lesson's sequence and content, including the same examples and the set questions given to students. It loses a point because the interactive discovery element (students identifying properties from physical shapes, deriving volume formulae) is replaced by direct exposition, slightly reducing fidelity to how the lesson was actually taught.

- `C1` The pack closely reflects the lesson's sequence, emphasis, and method. It loses one point because the sphere's properties receive only a brief summary relative to the cylinder, and the pack does not reproduce the teacher's in-class set questions about the cylinder (e.g., 'How many faces does a cylinder have?') as prominently as the volume questions.
- `C1` The pack faithfully reflects the lesson sequence, derivation approach, and evaluation questions. It loses one point because the teacher's actual in-class set questions ('How many circular faces does a solid cylinder have?' and 'A cylinder has how many faces?') are replicated but the broader interactive discovery element cannot be fully reproduced in a take-home format, which is acceptable but a minor gap.
- `C2` The broad sequence and assessment questions align well with the taught lesson, but the addition of the triangular prism example represents content and depth beyond what was actually taught, weakening the alignment with the lesson's activities and scope.
- `C2` The pack mirrors the lesson's order, its key activities, and the depth signalled by the teacher's own questions. The inclusion of a triangular prism example goes slightly beyond what the lesson presentation describes, but the core taught content is faithfully represented.
- `C2` The pack closely mirrors the lesson sequence, activities, and depth. A minor gap is that the lesson explicitly had students derive the formula interactively, which the pack approximates but cannot fully replicate in a take-home format; otherwise alignment is strong.
- `C2` The pack faithfully reproduces the lesson's sequence and depth, and the worked examples align well with what the teacher demonstrated. The only minor misalignment is the inclusion of a triangular prism volume example, which was not part of the lesson plan or previous knowledge.
- `C2` The revision pack closely reflects the lesson's sequence, the teacher's activities (identifying parts, then formula derivation and worked examples), and the intended depth. The extra triangular prism example was not part of the taught lesson, which slightly reduces alignment.
- `C2` The revision pack mirrors the lesson's structure, emphasis, and activities closely, making it a faithful reflection of what was taught. The alignment is not perfect (5) because the lesson's instructional approach involved students identifying properties from displayed shapes and discussing them, which the pack cannot replicate, and there is an irrelevant triangular prism figure that was not part of the lesson.
- `C3` The pack closely follows the lesson's sequence and method, including the formula derivation approach and the same evaluation questions. The only minor gap is that the lesson's emphasis on students identifying properties from physical shapes through observation is replaced by direct exposition, but this is appropriate for a take-home format.
- `C3` The pack faithfully reflects the lesson's sequence and method, including the teacher's own questions and the worked-example style. A point is withheld because the pack introduces a triangular prism example that was not part of the lesson, slightly extending beyond what was taught.
- `C3` The pack closely follows the taught sequence, activities, and assessment questions. It adds a triangular prism extension and Example 2 (reverse volume), which go slightly beyond what the teacher's lesson plan shows, but the core alignment is strong.
- `C3` The pack closely reflects the lesson's sequence, emphasis, and expected depth; minor gaps exist (e.g., no explicit reference to students identifying properties themselves, as the lesson was interactive), but overall alignment is strong.
- `C3` The pack faithfully mirrors the lesson's sequence and activities, including the teacher's own set questions, but the unsolicited triangular prism worked example extends slightly beyond the taught lesson scope. Overall alignment is strong.
- `C3` The pack closely mirrors the lesson's order, activities, and evaluation questions, showing strong alignment. It docks one point because the triangular prism extension was not part of the taught lesson presentation and adds depth not shown in the plan.

Scope and conceptual correctness — C0 4.33, C1 4.83, C2 3.00, C3 3.50

- `C0` The mathematical content and shape properties are accurate throughout, and the pack stays within the expected lesson scope. The only minor issue is the treatment of cylinder 'edges' (curved edges), which is consistent internally but is a nuanced point that different textbooks handle slightly differently; this does not constitute a conceptual error at JSS1 level.
- `C0` No mathematical errors were found in any formula, example, or answer key. The scope is appropriately bounded to what the lesson planned to teach.
- `C0` The mathematics and geometry content are correct throughout, and the scope stays within JSS1 boundaries. The only minor issue is the intermediate unit labelling in worked examples (cm^2 before multiplying by the final dimension), which is technically sound but slightly non-standard for this level.
- `C0` No mathematical or conceptual errors were found; the scope stays within JSS1 expectations and the lesson boundary. A minor point is that describing a cylinder as having 'two curved edges' is a simplification that some curricula handle differently, but it is consistent with the lesson's own framing and is internally consistent throughout the pack.

- `C0` Every mathematical statement and calculation in the pack is correct, and the content stays strictly within the intended lesson boundary without introducing out-of-scope material or misconceptions.
- `C0` The conceptual content is accurate and stays within lesson scope. The only flaw is a minor unit-labelling error in one intermediate step of a worked example, which does not affect the final answer but could confuse a careful student.
- `C1` Every mathematical claim and calculation in the pack is correct, and the content stays strictly within the lesson boundary without introducing concepts beyond JSS1 level.
- `C1` The pack is conceptually accurate throughout and stays within the JSS1 lesson boundary. The treatment of curved edges is internally consistent even if terminology varies across textbooks, and no mathematical errors are present.
- `C1` There are no conceptual or computational errors, and the content neither under-shoots nor over-shoots the scope of the planned lesson.
- `C1` Every mathematical statement and property description in the pack is correct, and the content stays squarely within the boundary of the planned lesson without straying into adjacent topics or introducing errors.
- `C1` There are no conceptual or computational errors in the pack, and it remains entirely within the scope of the planned lesson without straying into unrelated territory.
- `C1` Every mathematical statement and numerical answer in the pack is correct, and the content stays precisely within the intended lesson scope without over-extending or under-delivering.
- `C2` All mathematical content presented is conceptually and numerically correct, but the pack exceeds the lesson boundary by introducing triangular prism volume, which was neither taught nor listed as an objective.
- `C2` The mathematics presented is correct throughout, but the triangular prism example extends beyond the lesson boundary without being flagged as enrichment, which slightly inflates the scope for a JSS1 take-home pack tied to specific objectives.
- `C2` All mathematics presented is correct and the scope is mostly appropriate, but the stray triangular-prism figure is an unexplained intrusion that falls outside the lesson boundary and could confuse JSS1 learners.
- `C2` The mathematics throughout is accurate and the conceptual descriptions are sound. However, introducing triangular prism volume takes the pack noticeably beyond the lesson boundary for JSS1 students at this point in the term, which lowers the scope score.
- `C2` The mathematics is accurate throughout, and the properties of cylinders and spheres are correctly stated. However, the triangular prism example introduces content outside the lesson boundary, and the non-standard use of 'curved edges' for a cylinder is a minor conceptual imprecision at JSS1 level.
- `C2` The mathematical content within scope is correct and well-explained, but the inclusion of an entirely irrelevant triangular prism figure is a meaningful scope error that could confuse students. This prevents a score higher than 3.
- `C3` The conceptual and mathematical content is accurate throughout, and the pack stays within lesson boundaries for the most part. However, the stray triangular prism figure description is a clear scope intrusion that should not appear in this lesson's revision material, preventing a top score.
- `C3` The mathematical content within the lesson scope is error-free, but the inclusion of the triangular prism example introduces material beyond the planned lesson boundary for JSS1 Week 5, which is a scope problem even though the prism calculation itself is correct.
- `C3` The mathematical content is correct throughout, but the inclusion of the triangular prism volume goes beyond the lesson's scope for a JSS1 Week 5 lesson on cubes, cuboids, cylinders, and spheres. This risks confusing students revising what was actually taught.
- `C3` The mathematical and conceptual content is correct throughout, but the out-of-scope triangular prism figure caption is a noticeable error that does not belong in this lesson and could confuse students.
- `C3` All mathematical content within scope is correct, and no conceptual errors are present. The only concern is the out-of-scope triangular prism example, which does not introduce errors but does exceed the lesson boundary.
- `C3` There are no mathematical or conceptual errors in the pack. The only issue is scope: the triangular prism content goes beyond the lesson boundary, which is a mild scope overreach for a JSS1 revision pack on this specific lesson.

Student-facing clarity and usefulness — C0 5.00, C1 5.00, C2 4.50, C3 4.83

- `C0` The pack is exceptionally well structured for independent home use: language is simple and direct, real-world contexts are familiar, step-by-step models scaffold the method clearly, and complete answer keys with explanations support self-correction — all highly appropriate for JSS1 learners working alone.
- `C0` The language is simple and age-appropriate, examples are concrete and relatable, and the layout supports independent revision effectively. This is well-suited as a take-home pack for JSS1 students.

- `C0` The pack is exceptionally well-structured for independent home use: language is simple and friendly, examples are concrete, and the worked solutions support self-checking. A JSS1 learner could work through this without teacher support.
- `C0` The pack is exceptionally well-structured for independent home use: language is accessible, progression is logical, and the answer section provides reasoning rather than bare numbers, which is exactly what a JSS1 student revising alone needs.
- `C0` The pack is exceptionally well-suited for independent home use: language is simple and age-appropriate, examples build from familiar objects, step-by-step solutions scaffold understanding, and the answer key supports autonomous learning without adult help.
- `C0` The pack is well-structured, uses simple language, and provides worked examples with answers, making it highly suitable as independent take-home revision material for JSS1 students.
- `C1` The pack is written in clear, simple English appropriate for JSS1 learners, uses everyday objects as concrete references, and provides enough scaffolding — including fully worked answers — to be genuinely useful as independent home revision material.
- `C1` The pack is exceptionally well-suited for independent home use: language is simple and direct, analogies are familiar, steps are explicit, and the answer key supports genuine self-study. A JSS1 student should be able to work through it without teacher assistance.
- `C1` The pack is written in plain, age-appropriate language with logical progression from explanation to worked examples to practice to answers, making it highly usable as a standalone take-home revision tool.
- `C1` The pack is exceptionally well-suited for independent home use: vocabulary is accessible, reasoning is explained in plain steps, real-world analogies ground each concept, and complete solutions let students verify and learn from their own attempts.
- `C1` The pack is well-structured for unsupported home use: clear language, familiar real-world objects, fully worked examples, and answer keys with explanations mean a JSS1 student can read, practise, and check their own work without a teacher present.
- `C1` The language is plain and age-appropriate, the structure (explanation → worked example → practice → answers) is well-suited to unsupported home study, and the answers section gives enough working for self-correction.
- `C2` The pack is well-structured, uses relatable language and clear worked examples, and the answer key with explanations makes it genuinely useful for a JSS1 student working alone at home; the minor redundancy and the out-of-scope prism section do not significantly harm usability.
- `C2` The pack is well-structured, uses clear language appropriate for JSS1, and gives students everything they need to revise independently — explanations, examples, practice, and answers with working.
- `C2` The pack is well-structured and student-friendly for independent home use; the only notable blemish is the unexplained triangular-prism figure, which could momentarily confuse learners, but it does not undermine the overall usability.
- `C2` The language is plain, the layout moves logically from explanation to example to practice to answers, and the self-contained structure is well-suited to independent home study at JSS1 level.
- `C2` This is an exceptionally clear and well-organised take-home pack. The language, layout, worked examples, and answer key collectively make it very accessible and self-sufficient for a JSS1 learner working alone.
- `C2` The pack is well-structured and largely accessible to JSS1 learners, with good use of examples, summaries, and self-check answers. The unexplained triangular prism figure is a clarity weakness for unsupported home use, preventing a top score.
- `C3` The pack is well-structured, uses simple language suited to JSS1 level, and the combination of explanations, worked examples, practice questions, and full answers makes it highly functional as independent take-home revision material.
- `C3` The language is accessible, the layout is logical, and the self-marking answer key makes this genuinely usable by a JSS1 student working alone. It is among the clearest take-home packs a student could receive for this topic.
- `C3` The pack is well-structured and written in clear, age-appropriate language that JSS1 students can follow independently. The unsignposted extension to triangular prisms slightly undermines its usefulness as a focused revision tool.
- `C3` The pack is exceptionally clear, well-structured, and student-friendly; a JSS1 learner working alone at home could follow it from explanation through practice to self-checking without needing additional support.
- `C3` The pack is well-structured, written in plain language appropriate for JSS1, and provides complete worked examples plus answers — making it highly usable as a standalone take-home revision resource.
- `C3` The language is simple, concrete, and well-structured for independent use by a JSS1 student. The layout moves logically from concept to example to practice to answer-check, making it highly useful as take-home revision material.